Machine Learning Using Python

Machine Learning (ML) is the process of learning patterns from data and using them to make predictions on new data. Instead of explicit programming, models **learn from historical datasets**.

ML is widely used in:

- Recommendation systems (Netflix, Spotify)
- Spam detection
- Medical diagnosis
- Fraud detection

Python Libraries Used:

- **NumPy** – numerical operations
- **Pandas** – data processing
- **Matplotlib / Seaborn** – visualization
- **Scikit-learn** – ML algorithms

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Supervised vs Unsupervised Learning

Supervised Learning

Uses **labeled data**. Learns mapping between **input (X) and output (Y)**.

Examples:

- Predicting if a patient is **diabetic**
- Email **spam classification**

Algorithms:

- Logistic Regression
- Decision Trees
- k-Nearest Neighbors (kNN)

Unsupervised Learning

Uses **unlabeled data**. Finds hidden patterns in data.

Examples:

- Grouping **customers by buying patterns**
- Music playlist creation with **similar songs**

Algorithm example:

- K-Means Clustering

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Logistic Regression (Linear Model for Classification)

Logistic Regression is a **linear model** used for **binary classification**. It predicts the **probability of a class**.

Uses **Sigmoid Function**:

Probability range: **0 to 1**

Example applications:

- Disease prediction
- Loan approval prediction
- Spam detection

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Decision Tree Classifier

Decision Tree is a **tree-based supervised learning algorithm**.

Structure:

- **Root Node** → starting decision
- **Intermediate Nodes** → conditions
- **Leaf Nodes** → final class label

Example: Predicting if a **customer will buy a product**.

Advantages:

→ Easy to interpret

→ Handles nonlinear relationships

→ Works with categorical data

Leaf Node

Final class label

Intermediate Node

Condition splits data

Root Node

Starting decision point

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K-Means Clustering and Model Evaluation

K-Means Clustering

Used for **unsupervised learning**. Groups similar data points into **k clusters**. **k** represents the **number of clusters**.

Applications:

- Customer segmentation
- Music recommendation systems
- Document clustering